

CO30137 - Perceptron drill (wos)

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End

Questions:

Q 1) A two-input perceptron uses $x_0 = 1$ for bias. The current weights are $w = (w_0, w_1, w_2) = (0, 1, -1)$ and $\eta = 0.1$. A training example $x = (x_1, x_2) = (1, 2)$ has the target $t = +1$. (AI-generated question)

- Write the perceptron training rule
- Compute the output o for this example. Is this misclassified?
- If it is, compute the new weights after update.

(d) Compute $w \cdot x$ again with the new weights. In which direction did it change? why?

- Answer:

$$a) w_i \leftarrow w_i + \Delta w_i \quad (\Delta w_i = \eta (t - o) x_i)$$

w_i : current weight	η : learning rate
Δw_i : delta weight	t : target output
x_i : current input	o : current output

b)

We have:

$$o(\vec{x}) = \text{sgn}(\vec{w} \cdot \vec{x})$$

where $\text{sgn}(\vec{w} \cdot \vec{x}) = \begin{cases} 1, & \text{if } \vec{w} \cdot \vec{x} > 0 \\ -1, & \text{otherwise.} \end{cases}$

Therefore, we have:

$$\vec{w} \cdot \vec{x} = w_0 + w_1 x_1 + w_2 x_2 = 0 + 1 - 2 = -1$$
$$\Rightarrow o(\vec{x}) = -1.$$

c) We have $\Delta w_i = \eta(t - o) x_i$

$$+ \eta = 0,1 \quad + o = -1$$

$$+ t = 1 \quad + x_i = (1, 2)$$

$$\Rightarrow \Delta \vec{w}_i = 0,1 \cdot [1 - (-1)] (1, 2)$$
$$= (0,2; 0,4)$$

We have perceptron training rule:

$$w_i \leftarrow w_i + \Delta w_i$$

$$\Rightarrow \vec{w}_i = (0, 1, -1) \text{ and } \Delta \vec{w}_i = (0, 0,2; 0,4)$$

$$\Rightarrow \vec{w}_{\text{new}} = (0, 1,2; -1,4)$$

d) New direction: $\vec{w}_{\text{new}} \cdot \vec{x} = 0,2 \cdot 1 + 0,4 \cdot 2 = 0,2 + 0,8 = 1$

Old direction: $\vec{w} \cdot \vec{x} = 1 \cdot 1 - 2 \cdot 1 = -1$

$\Rightarrow$ Increase the weight to obtain the target output

Q₂) What are the values of weight w_0 , w_1 and w_2 for perceptron whose decision surface is illustrated in figure 4.3? Assume the Surface crosses the x_1 axis at -1 , and the x_2 axis at 2

(Mitchell, Machine Learning (1997), Ex 4.1 (page 124))

Answer:

Assume that $w_0 = c$ (c is constant, $x_0 = 1$)
So the surface is a line (d)
$$\begin{cases} (d) \cap x_1 = 1 \\ (d) \cap x_2 = 2 \end{cases} \Rightarrow \begin{cases} \vec{w} \cdot \vec{x} = -2 \\ w_0 = 2 \end{cases}$$

Q₃). Consider two perceptions defined by the threshold expression
 $w_0 + w_1 \cdot x_1 + w_2 \cdot x_2 > 0$

Perception A has the weight values: $w_0 = 1, w_1 = 2, w_2 = 1$
and perception B has the weight values:
 $w_0 = 0, w_1 = 2, w_2 = 1$.

True or False? Perception A is more general than perception B (Mitchell, Machine Learning (1997), Ex. 4.3 (page 124))

Answer:

For perception A: $1 + 2x_1 + 1x_2 > 0$

$$\Leftrightarrow 2x_1 + x_2 > -1$$

For perception B: $0 + 2x_1 + x_2 > 0$

$$\Leftrightarrow 2x_1 + x_2 > 0$$

$\Rightarrow$ Perception B is more general than A

Q₄) Design a two-input perception that implements the boolean function $A \wedge \neg B$. Design a two-layer network of perceptions that implements $A \oplus B$.

Answer:

We have perception: $\hat{y} = \begin{cases} 1, & \text{if } w_1 x_1 + w_2 x_2 + w_0 \geq 0 \\ 0, & \text{otherwise} \end{cases}$ ($y = A \wedge \neg B$)

For inputs $(0,0) \rightarrow y=0 \rightarrow w_0 < 0$

$(0,1) \rightarrow y=0 \rightarrow w_2 + w_0 < 0$

$(1,0) \rightarrow y=1 \Rightarrow w_1 + w_0 \geq 0$

$(1,1) \rightarrow y=0 \rightarrow w_1 + w_2 + w_0 < 0$

Q5) Write the delta rule and compare it with perception rule.
a) Which output does each rule use to compute the update

b) Under what condition does each rule converge
(AI-generated questions)

Answer:

+) Perception learning rule: $w_i \leftarrow w_i + \Delta w_i$
where $\Delta w_i = \eta(t - 0)x_i$

+) Delta rule: $\sigma(\vec{x}) = \vec{w} \cdot \vec{x}$

a) Output: +) Perception learning rule: Discrete output
 $y = f(\text{net})$

+) Delta rule: linear output $\text{net} = w^T x + b$

b) - Perception learning rule:

Converges to zero error in a finite number of steps
only if data is linearly separable

- Delta rule:

Converges to the point of MMSE regardless of whether
data is linearly separable or not